

General documentation

Reactive power capability
Wind turbine class Nordex Delta4000

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1. Introduction

This document specifies the reactive power capability for wind turbines (WT) of class Nordex Delta4000 with the following outputs.

- 4000 kW WT
- 4380 kW WT
- 4500 kW WT
- 4800 kW WT

All specifications of active and reactive power relate to the reference point on the low voltage side of the WT transformer (see Fig. 1).

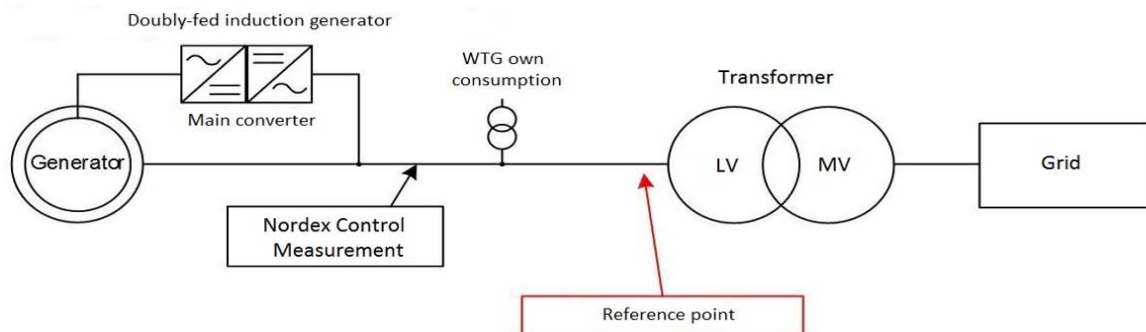


Fig. 1 Reference point of the active and reactive power

The maximum possible reactive power output of the WT depends on the conditions, see chapter 2.4 "Prerequisites", and on the respective active power output and thus also on the wind conditions.

2. Reactive power capability

2.1 Maximum reactive power capability

The Nordex Delta4000 wind turbine achieves the following reactive power values at the reference point and at nominal conditions (see Fig. 1).

Table 1: Maximum reactive power at the reference point (nominal conditions and rated output)

	Maximum reactive power (10-min average)							
	4000 kW WT		4380 kW WT		4500 kW WT		4800 kW WT	
	Inductive/ under- excited	Capacitati ve/over- excited	Inductive/ under- excited	Capacitati ve/over- excited	Inductive/ under- excited	Capacitati ve/over- excited	Inductive/ under- excited	Capacitati ve/over- excited
Q [kvar]	-2500	+2430	-2500	+2295	-2500	+2205	-2610	+2610
cos (φ)	0.848	0.855	0.869	0.886	0.875	0.898	0.878	0.878

2.2 Reactive power capability as a function of active power and grid voltage

The figures 2 - 5 below show the maximum possible reactive power for the turbine class Delta4000 as a function of the active power and grid voltage at the reference point (see Fig. 1).

The corresponding data are listed in tables 2-5. The following definitions apply

- -Q = inductive = under-excited
- +Q = capacitive = over-excited

It should be noted that the upper active power limit may be reduced dependent on the grid conditions if the required reactive power is very high. The maximum capacitive reactive power may also be limited by the grid conditions.

During this wind turbine standstill, e.g. if the wind speed is below the switch-on speed, no reactive power is consumed or fed in, and Q is 0. The "STATCOM function" option can be added to the standard reactive power range to provide reactive power also during WT standstill.

The voltage percentages refer to a nominal voltage U_n which can assume all nominal conditions for the voltage listed in chapter 2.4.

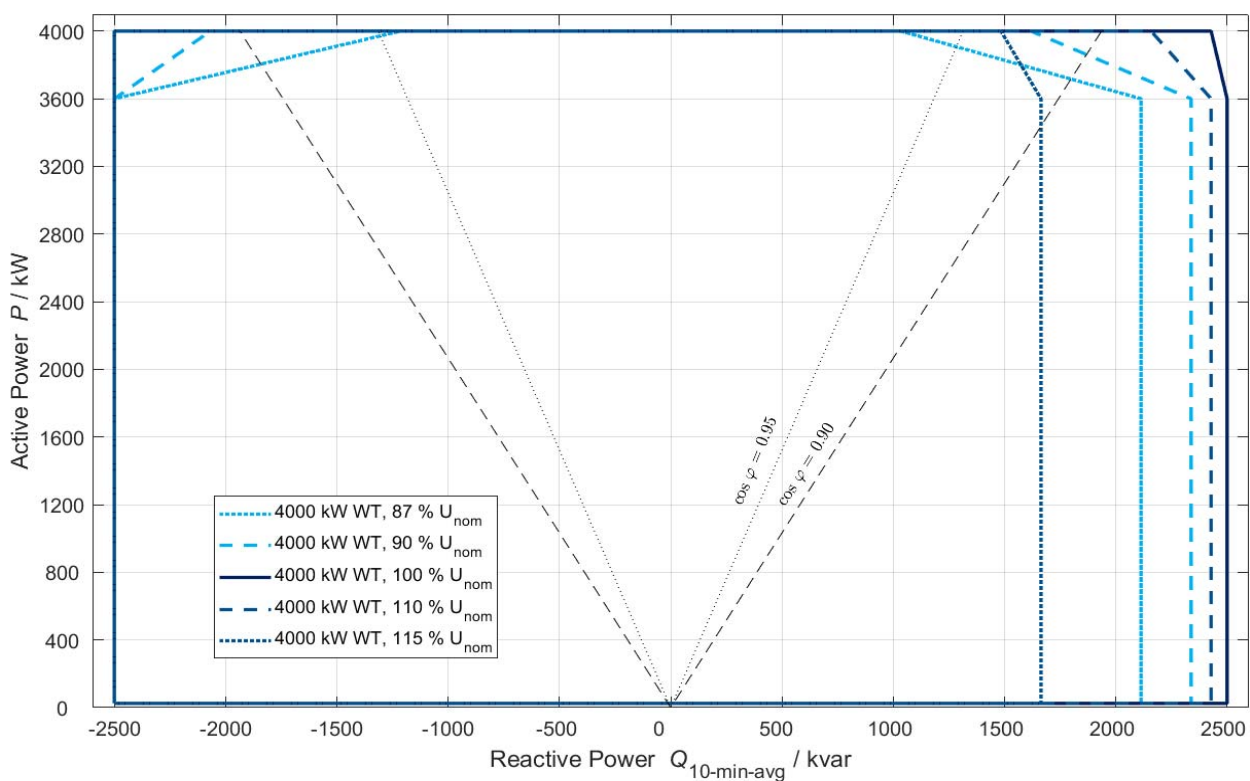


Fig. 2 4000 kW WT: Q-P diagram, see table 2

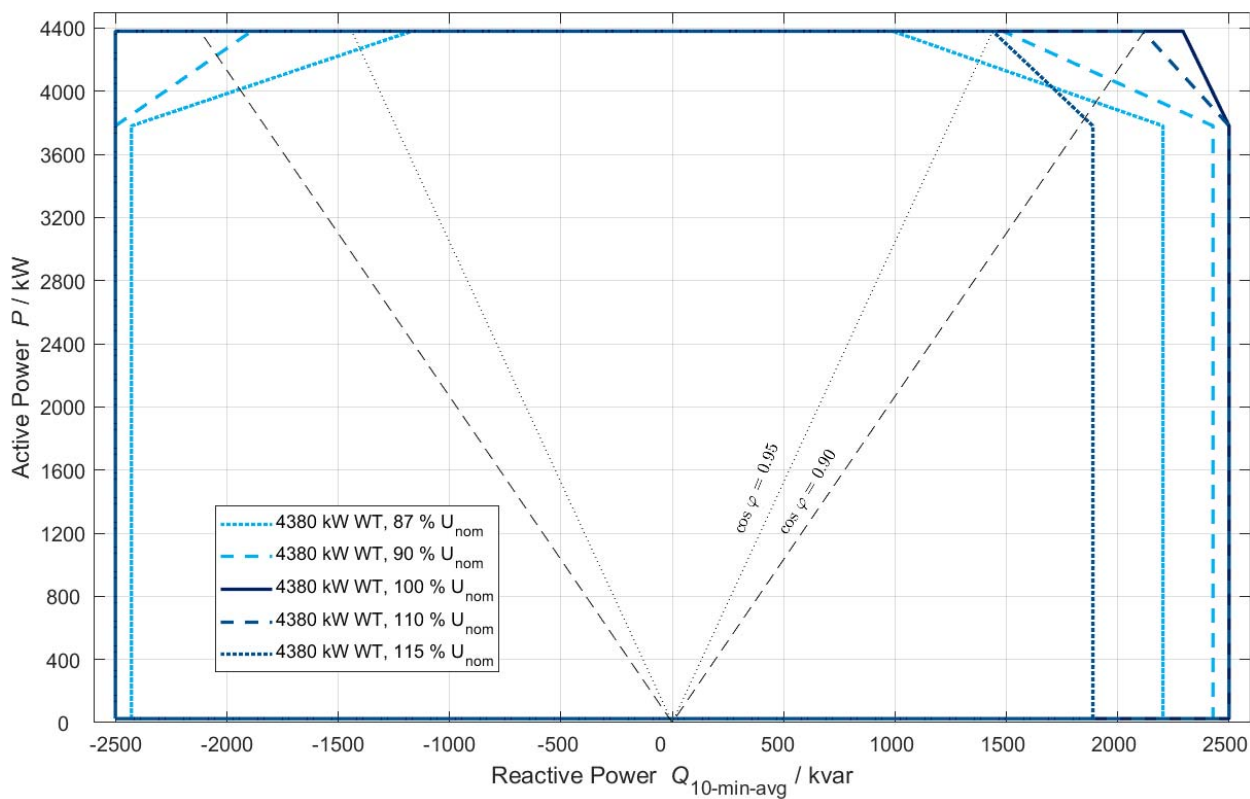


Fig. 3 4380 kW WT: Q-P diagram, see table 3

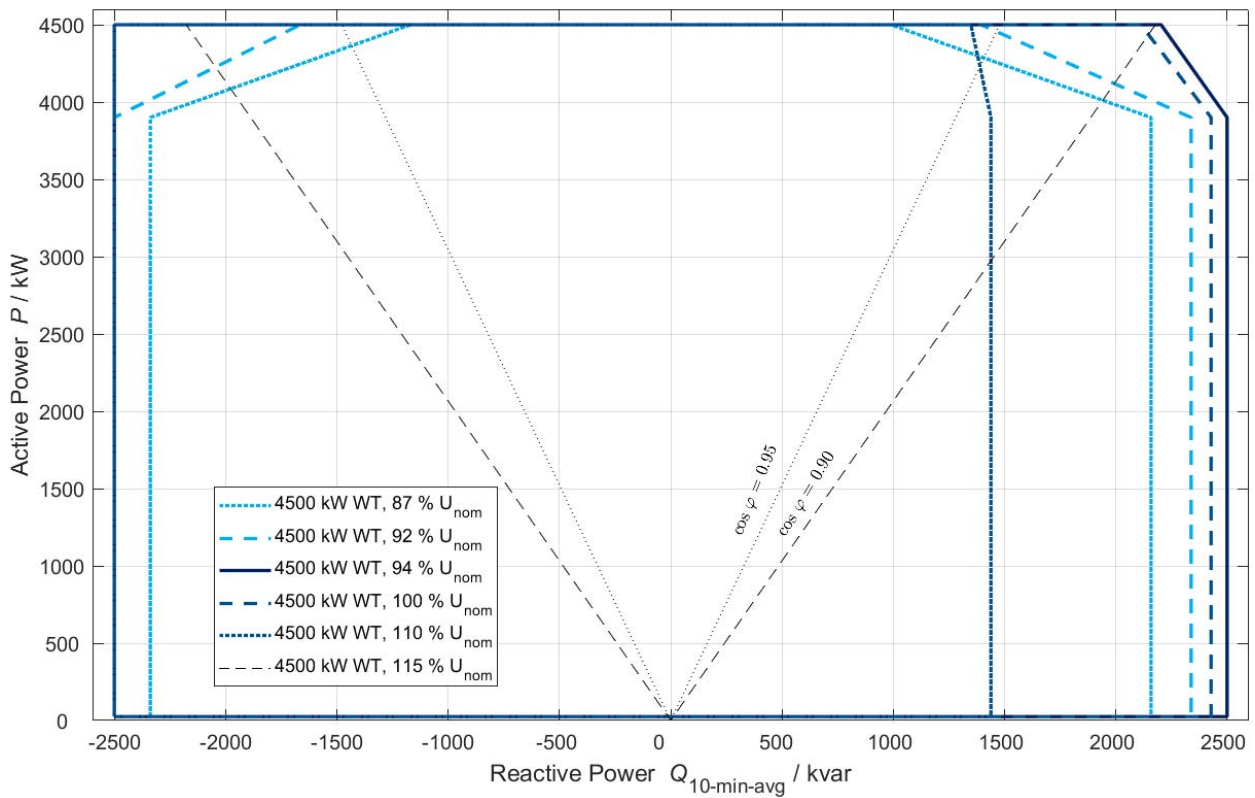


Fig. 4 4500 kW WT: Q-P diagram, see table 4

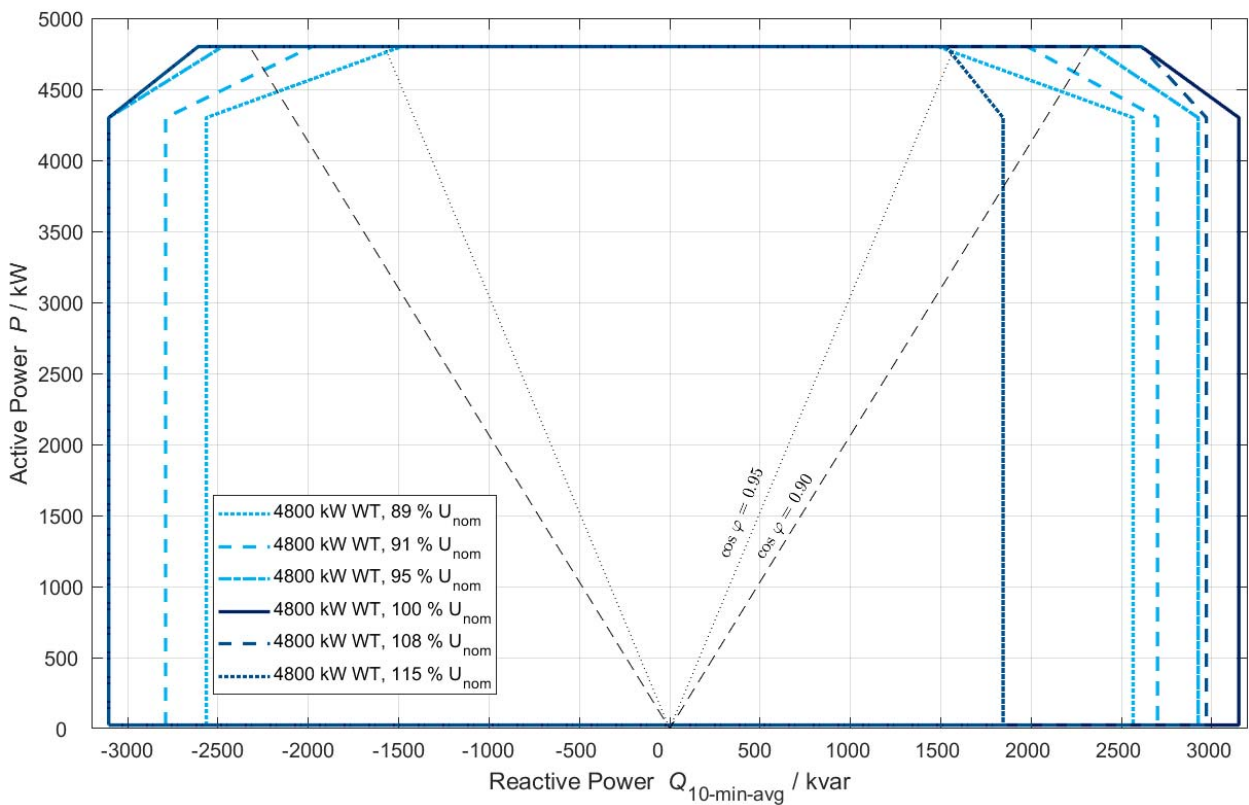


Fig. 5 4800 kW WT: Q-P diagram, see table 5

Table 2: Maximum possible reactive power of the 4000 kW WT in relation to active power and voltage at the reference point

Active power P [kW]		25	3600	4000
Maximum reactive power range (10-min average)	87 % U_n -Q...+Q [kvar]	-2500	-2500	-1215
		+2115	+2115	+1035
	90 % U_n -Q...+Q [kvar]	-2500	-2500	-2070
		+2340	+2340	+1620
	100 % U_n -Q...+Q [kvar]	-2500	-2500	-2500
		+2500	+2500	+2430
	110 % U_n -Q...+Q [kvar]	-2500	-2500	-2500
		+2430	+2430	+2160
	115 % U_n -Q...+Q [kvar]	-2500	-2500	-2500
		+1665	+1665	+1485

Table 3: Maximum possible reactive power of the 4380 kW WT in relation to active power and voltage at the reference point

Active power P [kW]		25	3780	4380
Maximum reactive power range (10-min average)	87 % U_n -Q...+Q [kvar]	-2430	-2430	-1170
		+2205	+2205	+990
	90 % U_n -Q...+Q [kvar]	-2500	-2500	-1890
		+2430	+2430	+1485
	100 % U_n -Q...+Q [kvar]	-2500	-2500	-2500
		+2500	+2500	+2295
	110 % U_n -Q...+Q [kvar]	-2500	-2500	-2500
		+2500	+2500	+2115
	115 % U_n -Q...+Q [kvar]	-2500	-2500	-2500
		+1890	+1890	+1440

Table 4: Maximum possible reactive power of the 4500 kW WT in relation to active power and voltage at the reference point

Active power P [kW]		25	3900	4500
Maximum reactive power range (10-min average)	87 % U_n -Q...+Q [kvar]	-2340	-2340	-1170
		+2160	+2160	+990
	90 % U_n -Q...+Q [kvar]	-2500	-2500	-1665
		+2340	+2340	+1395
	100 % U_n -Q...+Q [kvar]	-2500	-2500	-2500
		+2500	+2500	+2205
	110 % U_n -Q...+Q [kvar]	-2500	-2500	-2500
		+2430	+2430	+2115
	115 % U_n -Q...+Q [kvar]	-2500	-2500	-2500
		+1440	+1440	+1350

Table 5: Maximum possible reactive power of the 4800 kW WT in relation to active power and voltage at the reference point

Active power P [kW]		25	4300	4800
Maximum reactive power range (10-min average)	89 % U_n	-2565	-2565	-1485
	-Q...+Q [kvar]	+2565	+2565	+1485
	91 % U_n	-2790	-2790	-1980
	-Q...+Q [kvar]	+2700	+2700	+1980
	95 % U_n	-3105	-3105	-2475
	-Q...+Q [kvar]	+2925	+2925	+2340
	100 % U_n	-3105	-3105	-2610
	-Q...+Q [kvar]	+3150	+3150	+2610
	108 % U_n	-3105	-3105	-2610
	-Q...+Q [kvar]	+2970	+2970	+2610
	115 % U_n	-3105	-3105	-2610
	-Q...+Q [kvar]	+1845	+1845	+1530

2.3 Maximum reactive power capability on the medium-voltage side

The data at the reference point allows to calculate the Q/P diagrams on the medium-voltage side in consideration of the WT transformer data. Reactive power range and transformer losses may change depending on the transformer type. The data in Table 6 to Table 10 is based on expected type variations. For the corresponding reactive power range see fig. 6 - 9. For the calculations a grid frequency of 50 Hz has been assumed.

Table 6: Maximum reactive power on the medium-voltage side of the WT transformer (nominal conditions and rated output)

Maximum reactive power range (10-min average) under nominal conditions								
	4000 kW WT		4380 kW WT		4500 kW WT		4800 kW WT	
	Inductive/ under- excited	Capacitati ve/over- excited	Inductive/ under- excited	Capacitati ve/over- excited	Inductive/ under- excited	Capacitati ve/over- excited	Inductive/ under- excited	Capacitati ve/over- excited
Q [kvar]	-2747	+1845	-2717	+1755	-2670	+1690	-2983	+2088
cos (φ)	0.824	0.908	0.850	0.929	0.860	0.937	0.849	0.917

Table 7: Maximum possible reactive power of the 4000 kW WT in relation to active power and voltage on the medium-voltage side of the WT transformer

Active power P [kW]		25	3550	3950
Maximum reactive power range (10-min average)	90 % U_n -Q...+Q [kvar]	-1226	-1742	-1779
		+2270	+1966	+1526
	95 % U_n -Q...+Q [kvar]	-2641	-2887	-2559
		+2340	+2058	+1915
	100 % U_n -Q...+Q [kvar]	-2630	-2853	-2747
		+2322	+2063	+1845
	105 % U_n -Q...+Q [kvar]	-2622	-2822	-2869
		+2294	+2031	+1766
	110 % U_n -Q...+Q [kvar]	-2615	-2794	-2838
		+1788	+1524	+1320

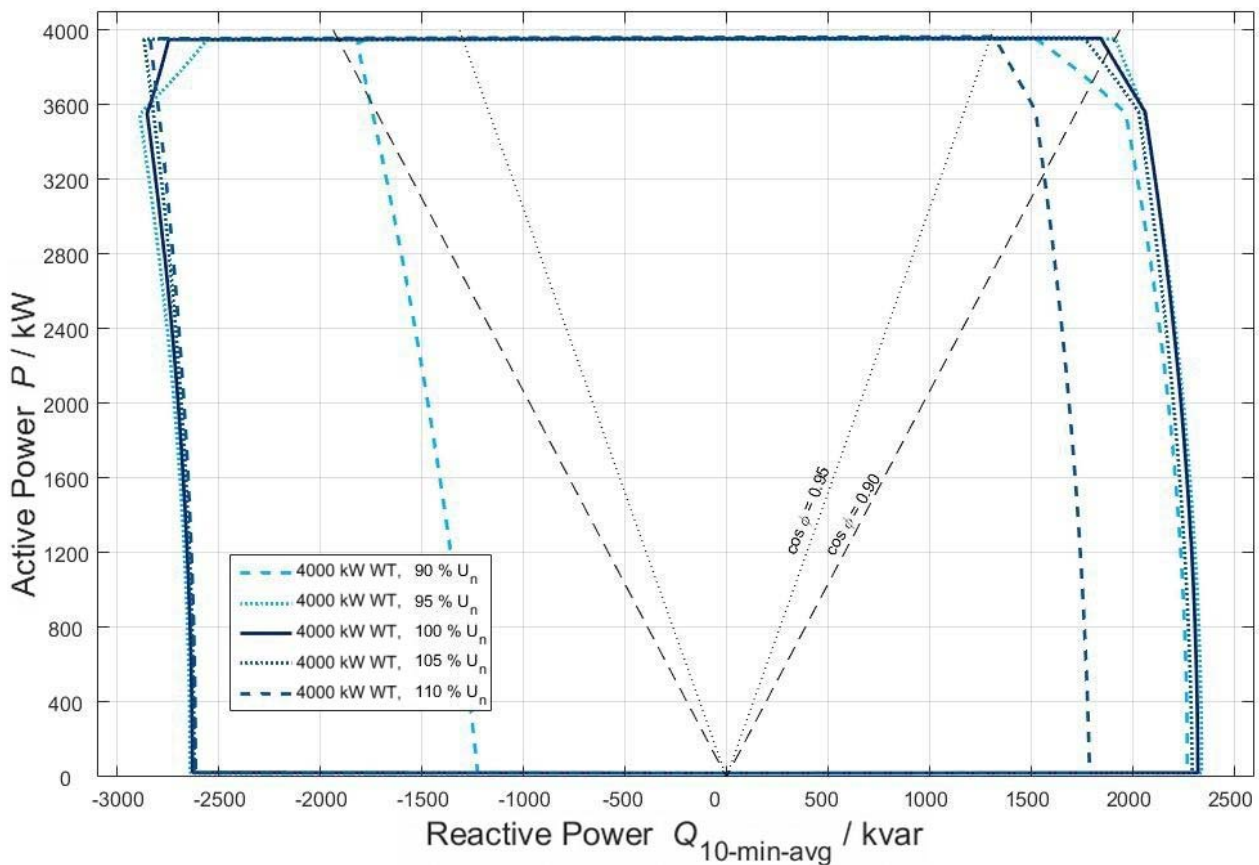


Fig. 6 4000 kW WT: Illustration of the Q/P diagrams on the medium-voltage side of the WT transformer (Refer to Table 7:)

Table 8: Maximum possible reactive power of the 4380 kW WT in relation to active power and voltage on the medium-voltage side of the WT transformer

Active power P [kW]		25	3730	4330
Maximum reactive power range (10-min average)	90 % U_n -Q...+Q [kvar]	-1376	-1866	-1814
		+2314	+2008	+1296
	95 % U_n -Q...+Q [kvar]	-2629	-2872	-2445
		+2354	+2076	+1766
	100 % U_n -Q...+Q [kvar]	-2620	-2837	-2717
		+2364	+2111	+1755
	105 % U_n -Q...+Q [kvar]	-2613	-2808	-2876
		+2371	+2136	+1711
	110 % U_n -Q...+Q [kvar]	-2608	-2783	-2845
		+1971	+1723	+1308

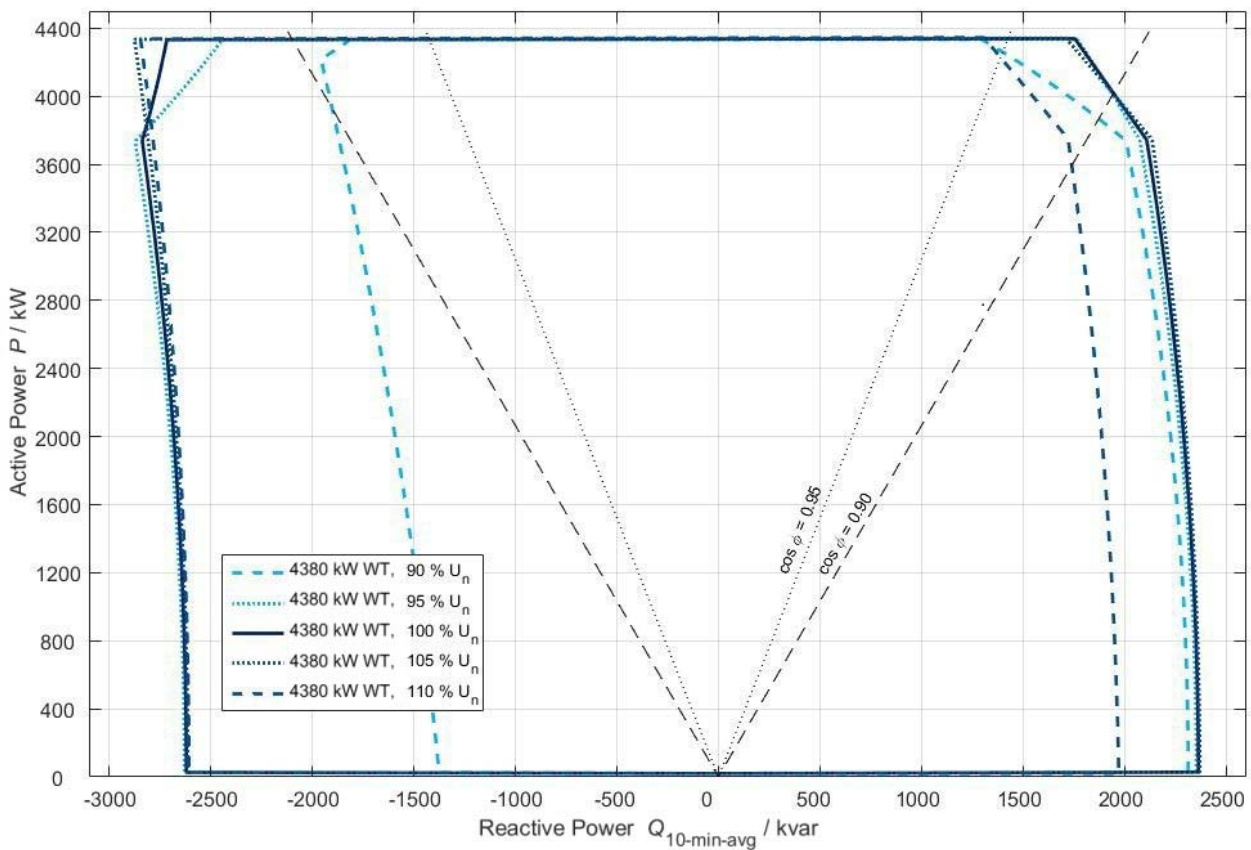


Fig. 7 4380 kW WT: Illustration of the Q/P diagrams on the medium-voltage side of the WT transformer (Refer to Table 8:)

Table 9: Maximum possible reactive power of the 4500 kW WT in relation to active power and voltage on the medium-voltage side of the WT transformer

Active power P [kW]		25	3850	4450
Maximum reactive power range (10-min average)	90 % U_n	-1376	-1888	-1803
	-Q...+Q [kvar]	+2273	+1949	+1170
	95 % U_n	-2629	-2887	-2299
	-Q...+Q [kvar]	+2350	+2055	+1652
	100 % U_n	-2620	-2852	-2670
	-Q...+Q [kvar]	+2334	+2061	+1686
	105 % U_n	-2613	-2821	-2891
	-Q...+Q [kvar]	+2311	+2061	+1686
	110 % U_n	-2608	-2795	-2859
	-Q...+Q [kvar]	+1730	+1452	+1246

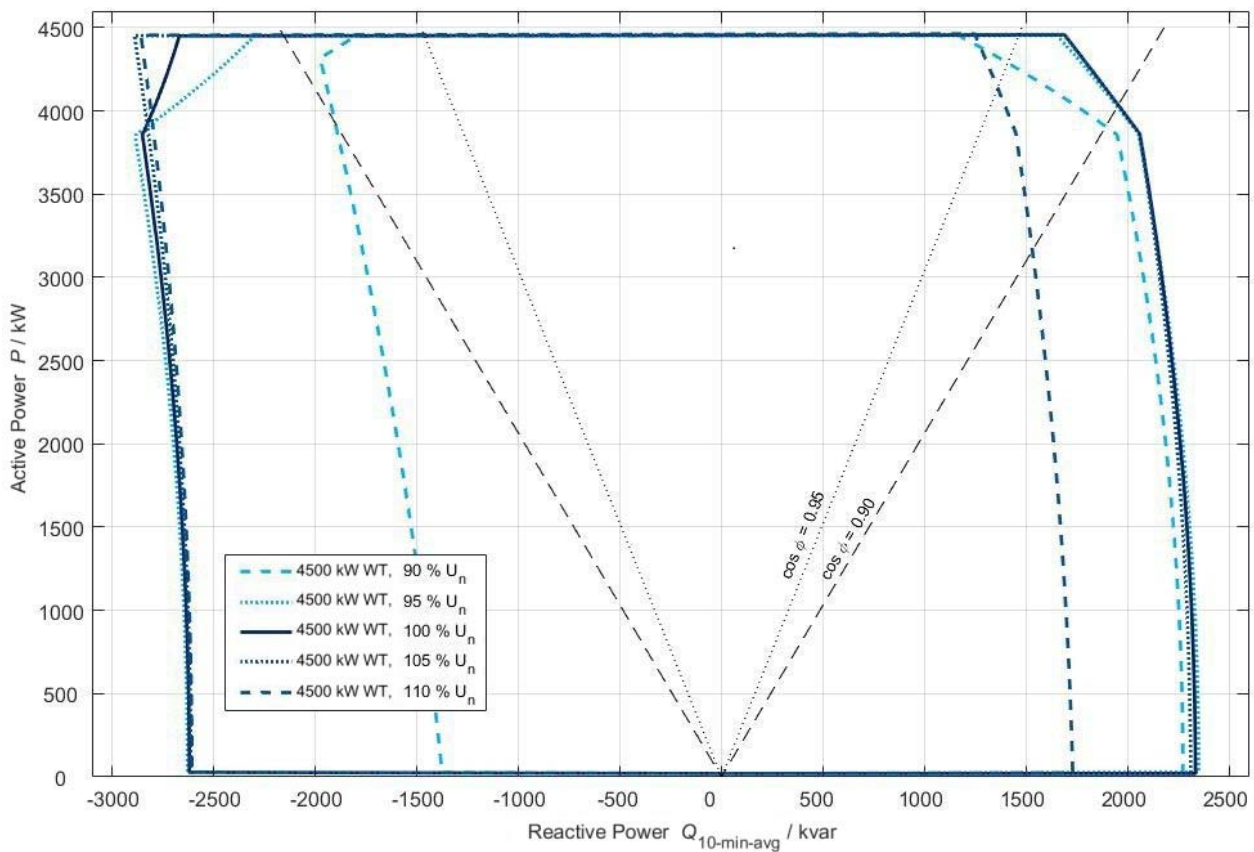


Fig. 8 4500 kW WT: Illustration of the Q/P diagrams on the medium-voltage side of the WT transformer (Refer to Table 9:)

Table 10: Maximum possible reactive power of the 4800 kW WT in relation to active power and voltage on the medium-voltage side of the WT transformer

Active power P [kW]		25	4250	4750
Maximum reactive power range (10-min average)	90 % U_n	-499	-1207	-1316
	-Q...+Q [kvar]	+2726	+2419	+1747
	95 % U_n	-2813	-3282	-2646
	-Q...+Q [kvar]	+2932	+2591	+2042
	100 % U_n	-3248	-3533	-2983
	-Q...+Q [kvar]	+2852	+2539	+2088
	105 % U_n	-3260	-3496	-3023
	-Q...+Q [kvar]	+2579	+2230	+1897
	110 % U_n	-3251	-3465	-2990
	-Q...+Q [kvar]	+2009	+1676	+1359

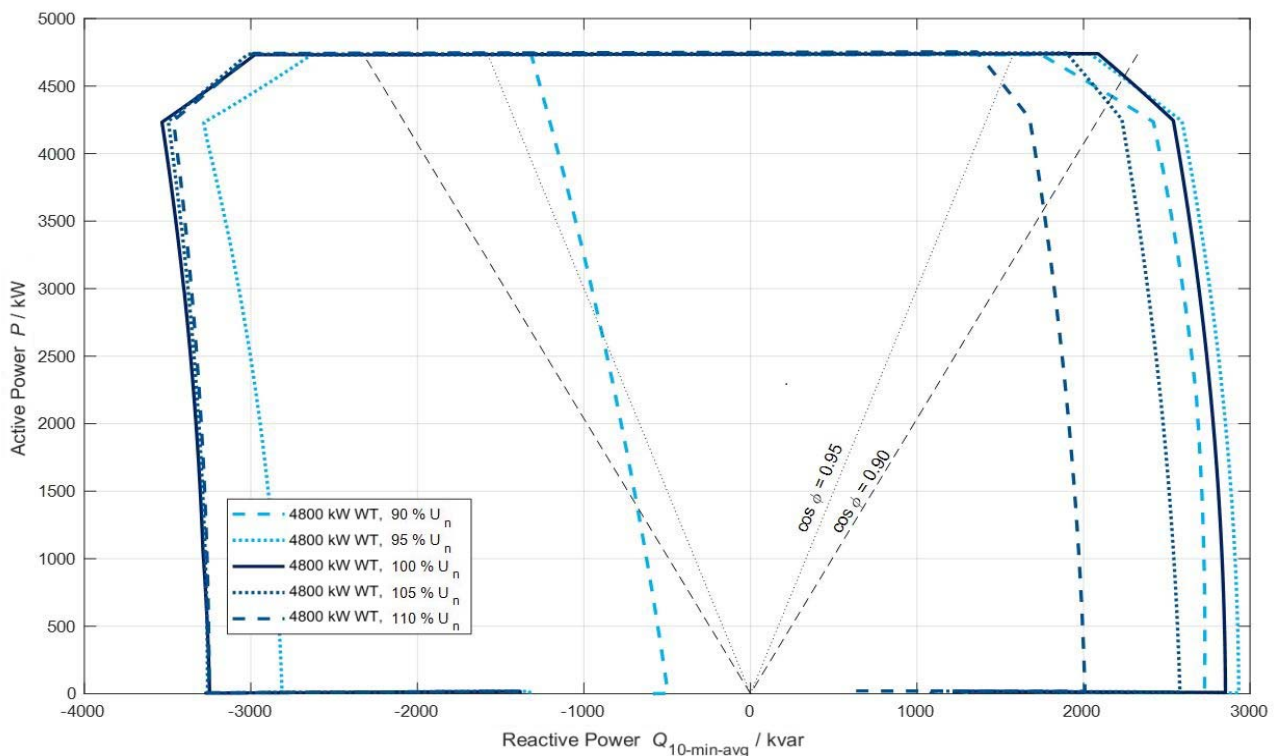


Fig. 9 4800 kW WT: Illustration of the Q/P diagrams on the medium-voltage side of the WT transformer (Refer to Table 10:)

2.4 Prerequisites

The data in chapter 2.1 to chapter 2.3 are based on the following framework conditions:

- Nominal conditions:
 - $U_{\text{reference point}} = 660 \text{ V}$ (4000 kW WT, 4380 kW WT, 4500 kW WT)
 - $U_{\text{reference point}} = 690 \text{ V}$ (4000 kW WT, 4380 kW WT, 4500 kW WT, 4800 kW WT)
 - $f_{\text{reference point}} = 50 \text{ Hz} / 60 \text{ Hz}$
- Grid conditions:
 - $f_{\text{reference point}} = 50 \text{ Hz} (\pm 3 \text{ Hz})$, $60 \text{ Hz} (-3 \text{ Hz} / +2,5 \text{ Hz})$
 - The simultaneous occurrence of extreme voltage and frequency values may cause a reduction of the maximum available active or reactive power.
- Ambient conditions:
 - Outside the specified ambient temperatures the maximum available active or reactive power may be reduced.
 - The data in chapter 2.1 to chapter 2.3 apply to the standard power curve. A noise-optimized and site-specific mode may require reductions of the maximum possible active or reactive power.
 - The WT must be in quasi-stationary operation. Strong gusts may require a temporary reduction of the maximum available reactive power. It may be assumed that the effects are not visible in the 10-min average values.

It should be noted that reference measuring results according to IEC 61400-21 contain a measuring uncertainty that might result in a measuring error of up to $\pm 2\%$ of the maximum reactive power value.

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